Credential Co-Generation and Verification Channel Orthogonalization for Digital Driver's Licenses in Heterogeneous Verifiable Credential Ecosystems: Formal Modeling of System Specifications and Feasibility Boundary Analysis

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***Abstract*—This paper presents a credential co-generation and verification-channel orthogonalization design for heterogeneous verifiable-credential ecosystems. At issuance, an authoritative issuer natively creates an SD-JWT VC and an ISO/IEC 18013-5 mDoc from the same attribute source and binds both to a deterministically computed credential reference. At presentation, remote verifiers use OpenID4VP with SD-JWT VC, whereas proximity verifiers use OpenID4VP proximity with mDoc; no format intermediary is introduced. A shared field-confirmation interface governs disclosure in both channels, while status tokens are bound to the verifier nonce and session identifier. System execution is formalized as a state machine, and source-consistency, disclosure-minimality, session-locality, and channel-independence invariants are defined and argued to be preserved over reachable states under the stated assumptions. Deployment feasibility is further characterized as a conjunction of jurisdictional and internal constraints, with analysis of decomposability, monotonicity, parameter sensitivity, and optimal-cost existence. Finally, applicability beyond digital driver's licenses is examined, together with scenarios in which dual issuance is unnecessary or unsuitable. The design is specification-consistent under its assumptions and conditionally deployable subject to jurisdiction-specific constraints.**

***Index Terms*—Digital driver's licenses, verifiable credentials, SD-JWT VC, mobile documents (mDoc), selective disclosure, OpenID4VP, credential status, formal modeling, deployment feasibility.**

I. Introduction

A. Problem Background

The core value of verifiable credentials lies not in digitizing paper documents, but in enabling a qualification or fact to be presented and verified in a manner that is verifiable, minimally disclosive, and reflective of real- time status. This goal imposes three simultaneous requirements on a credential system: the data presented must be independently traceable by the verifier to its issuance source; the scope of disclosure must be controllable field by field by the holder; and the credential status must reflect changes occurring after issuance. These three requirements correspond respectively to source traceability, disclosure minimization, and status timeliness. The absence of any one of them reduces a credential to "an image that can be screenshotted and forwarded."

Around this goal, two main technical lines currently exist. One is based on the W3C Verifiable Credentials data model, using JSON serialization and JWS signature containers, with OpenID4VCI for issuance and OpenID4VP for presentation; it has a mature ecosystem in remote verification scenarios. The other centers on mDoc as defined in ISO/IEC 18013-5, using CBOR encoding and COSE signature containers, designed for proximity and offline presentation, with IACA certificate chains as trust anchors; it has been deployed in multiple U.S. states, Japan, Australia, and the EU EUDI framework.

The two lines share almost no structure at the data representation layer: different encodings, different signature containers, different trust material publication methods, and different session management for presentation protocols. This divergence is not a temporary state of standards evolution, but the result of each ecosystem optimizing over time for different verification scenarios. Any credential system that attempts to serve both types of verifiers must directly address this divergence rather than assume it will converge in the short term.

B. Design Starting Point and Core Choice

The design starting point of this paper is a positive engineering choice: at the issuance stage, natively generate two credentials for the same attribute source separately, and at the presentation stage, introduce no format intermediary. This choice allows the two presentation paths to each complete native signature verification, so that the property "every attribute value seen by the verifier comes from some issuance" can be directly checked on the verifier side without relying on any intermediate component's self- declaration. This choice is not a negation of any existing practice, but an explicit value assignment to the design variable "at which stage the format coexistence problem is solved." Moving the problem forward to the issuance stage yields three properties that can be formally argued.

First, the credential reference is deterministically computed at issuance, and any presentation can be bound to a unique issuance record. This binding is directly checked locally by the verifier, so source consistency becomes a monotonic invariant over the reachable state set (Theorem 5.26 in Section V).

Second, the disclosure set is uniquely determined by the field confirmation interface, and both channels share the same interface output. Accordingly, cross-channel disclosure consistency degenerates into an identity rather than a lemma requiring independent proof (Theorem 5.27 in Section V).

Third, the status token's signature covers the verifier nonce and session identifier, making cross-session reuse an event directly excludable at the signature verification layer; thus session locality is directly given by the conjunctive conditions of the verification predicate (Lemma 5.15 and Theorem 5.26 in Section V).

The cost of this choice is that the issuer must maintain two signature processes and two key lifecycles. This paper does not conceal this cost; instead, Section VI formalizes it as a system-controllable constraint in the feasibility decision and provides an existence proof for the optimal deployment cost solution (Theorem 6.11).

C. Contributions

The work of this paper revolves around a core question: in an ecosystem with fragmented credential formats, can a credential simultaneously maintain source traceability, disclosure controllability, and status timeliness across both remote and proximity verification paths, without relying on any intermediate component that is neither trusted by users nor audited by verifiers?

Around this question, this paper uses a digital driver's license as the first complete instance, providing a complete account of roles, timing, status management, and scope exclusions; abstracts system execution into a state machine and event traces, defines system invariants, and provides rigorous arguments in the form of auxiliary lemmas, a main theorem, and corollaries; formalizes deployment feasibility as a conjunction of constraint predicates and proves structural properties of the feasible region (decomposability, monotonicity, optimality existence, parameter sensitivity); and further tests the design in scenarios beyond digital driver's licenses, providing item-by-item checks of applicability and inapplicability counterexamples, thereby clarifying its applicability boundary rather than making generalized claims.

The common thrust of the above work is: the value of a technical solution lies not in eliminating ecosystem divergence, but in making the consequences of divergence computable and auditable.

D. Organization

Section II covers the background and standards maturity of the two ecosystems. Section III presents the abstract model and applicability criteria. Section IV develops the complete design using the digital driver's license as an instance. Section V presents formal modeling of system specifications, including auxiliary lemmas, the main theorem, and corollaries. Section VI presents the formalization of feasibility boundaries, including the feasible region characterization theorem, structural properties, and optimization analysis. Section VII analyzes cross-domain applicability, with detailed analysis in Appendix A. Section VIII discusses argument boundaries and limitations. Section IX concludes.

II. Background and Related Work

A. W3C VC Ecosystem

The W3C Verifiable Credentials data model organizes credentials in a three-party structure of "issuer– holder–verifier." A credential is an assertion by the issuer about a set of claims, and the holder proves the legitimacy of the presentation by possessing key material bound to the credential. The data model layer does not prescribe encoding or signature algorithms; in practice, two main encoding paths exist: JSON-LD with Data Integrity suites, and JWT with JWS containers.

The shortcoming of the W3C VC data model is that it does not natively support field-level selective disclosure. The JSON-LD path requires derived proof suites for field-level disclosure; the JWT path uses SD- JWT's "salted hash + selective release" mechanism to achieve the same goal. SD-JWT VC has therefore become the profile in the W3C VC ecosystem most aligned with selective disclosure requirements, and is the encoding chosen in this paper.

For issuance and presentation protocols, OpenID4VCI handles credential issuance and OpenID4VP handles presentation requests and responses. Their value lies in decoupling "credential format" from "transport protocol": the same presentation protocol can carry multiple credential formats, which provides the protocol-layer basis for the channel orthogonalization in this paper.

B. mDoc Ecosystem

The mDL defined in ISO/IEC 18013-5 is a mobile driver's license profile designed specifically for proximity presentation. Its credential format is mDoc, using CBOR encoding and $\text{COSE}_{\text{Sign1}}$ signature containers, with fields organized by namespace. Each attribute element exists independently, so field-by-field disclosure is an inherent property of the data structure rather than an added mechanism. Proximity presentation consists of two phases–device engagement and device retrieval–and is designed to complete device-to- device credential exchange without network connectivity.

The trust anchor for mDoc is the IACA certificate chain. Each country establishes an IACA root and achieves cross-border mutual recognition through cross-certification. The advantage of this system is clear and auditable trust relationships; the disadvantage is a closed ecosystem with high barriers for new issuers, and CBOR/COSE is unfriendly to Web developers.

The ISO/IEC 23220 series extends the mDoc architecture from driver's licenses to general mobile eIDs, defining security mechanisms for issuance, holding, and presentation phases. Its transport layer is compatible with 18013-5, but current production deployment cases are extremely few, and the verifier-side ecosystem has not yet formed.

C. Standards Maturity and Interoperability Status

The three standards involved in this paper are at different maturity stages. W3C VC 2.0 has become a Recommendation, SD-JWT VC is still an IETF draft, OpenID4VP has been formally released, and ISO/IEC 23220-4 is a newly published standard in 2026. Large-scale interoperability testing among the three has not yet been publicly completed. The impact of this status on this paper is: the system assumptions on which the formal parts depend must be explicitly listed (see Section V-A), and the capability claims in the engineering parts must converge on stable facts rather than expected behavior in drafts.

D. Positioning of This Paper

This paper does not argue that one ecosystem should prevail, nor does it attempt to define cross-ecosystem trust mutual recognition rules. This paper addresses a narrower but more fundamental question: can a credential system that must serve both remote and proximity verifiers, without introducing a format intermediary, enable each path to satisfy the three requirements of source traceability, disclosure minimization, and status timeliness? Section III presents the abstract model, Section IV develops the driver's license instance, Sections V and VI provide formal arguments, and Section VII and Appendix A analyze cross- domain applicability.

III. Abstract Model

A. Applicability Criteria

The design in this paper is not necessary for all credential types. It targets a class of verifiable credentials that simultaneously satisfy the following criteria:

P1 (Single and Authoritative Attribute Source). Credential attributes come from an identifiable authoritative issuer, and the issuer bears responsibility for attribute authenticity.

P2 (Changeable Status). After issuance, the credential has status changes such as validity period, revocation, and suspension, and the verifier needs to perceive the current status rather than the status at issuance.

P3 (Disclosure Minimization Required). Different verifiers require different fields; full disclosure violates the minimization principle and requires field-by-field control.

P4 (Coexisting Dual-Channel Verification). The same credential must serve both remote verifiers and proximity verifiers.

P5 (Coexisting Format Ecosystems). The target market simultaneously contains W3C VC and mDoc verifiers, and neither side accepts native verification of the other's format.

P1 to P3 determine "whether a verifiable credential is needed"; P4 determines "whether dual channels are needed"; P5 determines "whether co-generation is needed." The applicability domain of the design in this paper is jointly defined by the above criteria, and Section VII will check them scenario by scenario.

B. Roles and System Assumptions

The system contains three roles. The issuer is an authoritative issuance entity or its authorized technical entity, responsible for credential generation and status publication, and maintains issuance records. The wallet is controlled by the holder and is responsible for credential storage and presentation. The verifier is a remote system or proximity terminal responsible for initiating presentation requests and verifying responses.

There are four system assumptions, corresponding to the modeling in Section V-A.

S1 (Issuance Record Integrity). The issuer saves a record for each issuance, containing the credential reference ref and the attribute set A of this issuance. Records cannot be modified after the fact. S2 (Reference Function Determinism): The credential reference is computed by a deterministic function from the attribute set and issuance metadata; the same attribute set in the same issuance produces the same reference, and different attribute sets do not produce the same reference.

S3 (Device Key Exclusivity). The wallet device key is exclusively held by the system, and third parties cannot obtain its use right without triggering a security event.

S4 (Verification Material Integrity). The channel through which the verifier obtains the issuer's verification material has integrity protection, and adversaries cannot replace the material without being detected.

All four assumptions are verifiable conditions at the system operation level and do not involve probabilistic statements. S1 and S4 are deployment requirements, S2 is a function selection requirement, and S3 is a device capability requirement.

C. Credential Co-Generation

The issuer, in a single generation operation, produces two credentials for the same attribute source.

SD-JWT VC. Uses a JWS signature container, with the cnf claim pointing to the wallet device public key, and fields containing all attributes required by the credential. Undisclosed fields are retained as placeholders during presentation, thereby structurally supporting field-by-field disclosure.

mDoc. Uses a $\text{COSE}_{\text{Sign1}}$ signature container, with DeviceKeyInfo in the MSO pointing to the wallet device public key, and attribute elements co-sourced with the SD-JWT VC.

The two credentials share the same credential reference ref, which is computed by the deterministic function described in S2 from the attribute set and issuance metadata. The role of ref is to provide verifiable evidence that "the two credentials point to the same issuance," not to provide evidence that "the holders of the two credentials are the same person"–the latter is a scope exclusion stated in Section IV-C.

The issuer separately publishes a JWKS endpoint and an mDoc issuer public key endpoint; the two are independently maintained and independently rotated, with no association claims. Verifiers select the corresponding material according to the format received and do not require a mapping between the two materials.

Co-generation makes the two credentials each complete at generation time and each traceable to the same issuance record, with no format transfer during presentation. The cost is that the issuer must maintain two signature processes and two key lifecycles.

D. Verification Channel Orthogonalization

The remote channel uses a combination of OpenID4VP and SD-JWT VC. The verifier sends a presentation request declaring the requested field set D_{req} and a nonce non_{V}. The wallet parses the request, presents D_{req} on the field confirmation interface, and after user confirmation obtains $\text{D}_{\text{sel}} \subseteq \text{D}_{\text{req}}$. The wallet releases the corresponding disclosed claims and signs the disclosed content and non_{V} with the device key, returning over HTTPS.

The proximity channel uses a combination of OpenID4VP proximity and mDoc. The verifier displays a QR code carrying the startup information for a local session; the wallet scans it and acts as a local server to establish an HTTPS session.

The verification device connects and follows the standard OpenID4VP flow, and the wallet returns an mDoc presentation. This design reuses the protocol stack and session management of the remote channel and does not depend on the transport layer implementation of ISO/IEC 23220. mDoc here serves only as a data-bearing format; its signature and device authentication are still verified according to native rules.

The meaning of orthogonalization is: the verification logic of the two channels is mutually independent, and the correctness of either channel can be established independently without cross-channel consistency assumptions. This is reflected in Invariant I4 and the corresponding theorem in Section V as "the proof does not cross channel boundaries."

Both channels share the same field confirmation interface. The user's choice determines which SD-JWT claims are released in the remote channel and which attribute elements are included in the mDoc in the proximity channel. The interface behavior is consistent, the underlying encodings differ, and the user does not need to understand the encoding differences.

E. Status Management

Status management uses the IETF Token Status List. The status token contains only the credential reference ref and the status bit, with no holder identifier. The token's signature covers the verifier nonce non_V and the presentation session identifier sid, so one presentation corresponds to one token and cannot be reused across sessions. The token contains no persistent identifier across verifiers, thereby structurally suppressing cross-verifier correlation.

The status token is transmitted together with the SD-JWT VC or mDoc. In offline scenarios, a short-lived token allows the verifier to make an immediate judgment at presentation; after reconnecting, the issuer status endpoint can be queried separately. The token validity window and offline acceptance rules are defined by system parameters and do not depend on the verifier's discretion.

IV. Digital Driver's License Instance

A. Roles and Attributes

In the digital driver's license scenario, the issuer is the driver's license administration authority or its authorized technical entity; the wallet is the license holder's mobile application; verifiers are divided into remote (rental car web pages, insurance underwriting systems) and proximity (traffic police handheld terminals, hotel front desks). The attribute set A includes name, date of birth, permitted vehicle categories, validity period, document number, and document status.

This scenario satisfies the above criteria: P1 holds (the traffic management authority is the authoritative source); P2 holds (revocation, suspension, expiration); P3 holds (rental car companies do not need address, hotels do not need permitted vehicle categories); P4 holds (remote and proximity verification coexist); P5 holds (W3C VC verifiers and mDoc verifiers coexist in the target market).

B. Interaction Timing

Issuance timing. Verify holder identity → issuer computes attribute set and ref → generate SD-JWT VC and mDoc in parallel → wallet receives and writes to secure storage → wallet returns receipt confirmation → issuer records issuance record (the two credentials have the same ref as evidence of completed co- generation). Remote presentation timing: Verifier generates non_V and D_{req} → wallet parses request and presents field confirmation interface → user confirms D_{sel} → wallet generates disclosed claims and device signature → issuer status endpoint issues a session-bound status token → wallet returns presentation → verifier verifies source traceability, status token, and session binding.

Proximity presentation timing. Verifier displays QR code → wallet establishes local session → verifier sends request containing non_V → user confirms D_{sel} → wallet generates mDoc presentation and device authentication signature → status token attached → verifier verifies.

Status change timing. Issuer updates status bit → status endpoint generates a new token template → wallet obtains a token bound to the current session at the next presentation → verifier perceives the status change at presentation.

C. Design Trade-offs and Scope Exclusions

The main trade-off of co-generation relative to single issuance is: at the cost of the issuer's dual signature infrastructure, it gains the ability for the two presentation paths to each complete native signature verification. This paper considers this exchange reasonable, because the former is a measurable engineering cost, while the latter determines whether source traceability can be directly checked by the verifier.

The scheme explicitly excludes three items and does not design mechanisms for them: cross-format holder public key equivalence; unified trust anchor mapping between JWKS and mDoc issuer public keys; and equivalence declarations with the ISO/IEC 18013-5 IACA trust chain. These three are cross-ecosystem governance issues beyond the technical design scope of this paper; their impact enters the feasibility decision in Section VI as external parameters and is not promised at the design layer.

V. Formal Modeling of System Specifications and Invariant

Arguments This chapter abstracts the design of Section III into a state machine and event traces, defines system invariants, and provides rigorous arguments in the form of auxiliary lemmas, a main theorem, and corollaries. The arguments do not rely on probabilistic statements, but on induction over the reachable state set, analysis of state transition function properties, and algebraic characterization of the partial order structure of disclosure sets.

A. State Machine Formalization

Definition 5.1 (System State): A system state is defined as a four-tuple

$$s = (R, K, T, E)$$

where R is the set of issuance records; K is the set of credentials held by the wallet; T is the set of active status tokens; and E is the event history, ordered by time.

Definition 5.2 (Initial State): The initial system state is $s_0 = (\emptyset, \emptyset, \emptyset, \langle\rangle)$.

Definition 5.3 (Event Types): The system supports four types of events:
`issue(r)`: issuance record r is added to R, and the corresponding credential is added to K;
`show(e)`: presentation event e is added to E; token(T) : status token T is added to T;
`update(ref, bit)`: updates the status bit of the credential corresponding to ref.

Definition 5.4 (State Transition Function): The state transition function δ: S × Event → S is defined as:

$$\delta((R, K, T, E), issue(r)) = (R \cup \{r\}, K \cup \kappa(r), T, E)$$

$$\delta((R, K, T, E), show(e)) = (R, K, T, E \cdot \langle e \rangle)$$

$$\delta((R, K, T, E), token(T')) = (R, K, T \cup \{T'\}, E)$$

$$\delta((R, K, T, E), update(ref, bit)) = (R', K, T, E)$$

where R' is the result of updating the status bit of the record corresponding to ref in R.

Definition 5.5 (Reachable State Set): The reachable state set is defined as

$$\text{Reach}(s_0) = \{s : \exists \sigma = \langle ev_1, \ldots, ev_n \rangle, \delta^*(s_0, \sigma) = s\}$$

where δ^* is the natural extension of δ over event sequences.

Definition 5.6 (Issuance Record): An issuance corresponds to a record r = (ref, A, t_{issue}), where ref is the credential reference, A is the attribute set, and t_{issue} is the issuance time.

Definition 5.7 (Presentation Event): A presentation corresponds to an event e = (ref, D_{sel}, ch, non_V, sid, t), where ch ∈ {remote, proximity} is the channel type, non_V is the verifier nonce, and sid is the session identifier.

Definition 5.8 (Status Token): A status token is a tuple T = (ref, $status_{bit}$, non_V, sid, t), where $status_{bit}$ ∈ {valid, suspended, revoked}.

Definition 5.9 (Verification Decision Function): The verification decision is a predicate Verify(e, R, T) → {accept, reject}, with decision rules as the conjunction of the following conditions:

1) ∃ r ∈ R such that e.ref = r.ref
2) $e.D_{sel} \subseteq r.A$
3) $T.ref = e.ref \wedge T.non_V = e.non_V \wedge T.sid = e.sid$
4) e.t and T.t are both within the validity window
5) device signature and issuer signature verification pass.

B. Algebraic Structure and Lattice Properties of Disclosure Sets

Definition 5.10 (Lattice Structure of Disclosure Sets): The power set of the attribute universe U is used as the carrier of disclosure sets, with partial order ≤ defined as set inclusion ⊆, meet ∧ as set intersection ∩, and join ∨ as set union ∪. Then (P(U), ⊆, ∩, ∪) forms a complete Boolean lattice with minimum element ∅, maximum element U, and complement operation U \{ } ·.

Lemma 5.11 (Distributive Law of Lattice Operations): For any D_1, D_2, D_3 ∈ P(U):

$$D_1 \wedge (D_2 \vee D_3) = (D_1 \wedge D_2) \vee (D_1 \wedge D_3)$$

Proof. Distributive law of set intersection and union, a standard result. □

Lemma 5.12 (Monotonic Contraction of Disclosure Sets): For any presentation event e and requested field set D_{req}, if $D_{sel} \subseteq D_{req}$, then $D_{sel} \wedge D_{req} = D_{sel}$

That is, the disclosure set confirmed by the user is not greater than the requested set under the lattice order.

Proof. From $D_{sel} \subseteq D_{req}$ directly follows $D_{sel} \cap D_{req} = D_{sel}$. □

Lemma 5.13 (Reference Determinism Lemma): By S2, there exists a deterministic function F_{ref}: P(U) × M → REF such that for all r ∈ R, r.ref = F_{ref}(r.A, r.meta). Furthermore, F_{ref} satisfies injectivity:

$$F_{ref}(A, meta) = F_{ref}(A', meta') \implies A = A' \wedge meta = meta'$$

Proof. By the definition of S2, F_{ref} is a deterministic function and different attribute sets do not produce the same reference. Injectivity follows directly from S2. □

Lemma 5.14 (Interface Determinism Lemma): There exists a deterministic function F_{UI}: SID → P(U) such that for any presentation event e, $e.D_{sel} = F_{UI}(e.sid)$.

Proof. By Section III-D, the presentation generation functions of both channels take D_{sel} as their sole input; the interface produces only D_{sel}, not encodings. Therefore $e.D_{sel}$ equals the set output by the interface on sid, and the existence of F_{UI} follows directly from the determinism of the interface. □

Lemma 5.15 (Token Binding Lemma): If status token T is generated by presentation event e', then for any presentation event e:

$$\text{ValidToken}(T, e) = 1 \implies e.sid = e'.sid \wedge e.non_V = T.non_V$$

Proof. By Definition 5.9 condition (iii), ValidToken(T, e) = 1 requires T.sid = e.sid and $T.non_V = e.non_V$. Since T is generated by e', T.sid = e'.sid, hence e.sid = e'.sid. □

Lemma 5.16 (Embedding Property of Disclosure Sets): For any presentation event e, if Verify(e, R, T) = accept, then there exists r ∈ R such that

$$F_{UI}(e.sid) \subseteq r.A \subseteq U$$

Proof. By Verify condition (ii), $e.D_{sel} \subseteq r.A$; by Lemma 5.14, $e.D_{sel} = F_{UI}(e.sid)$; since A is an attribute set, $A \subseteq U$. Therefore $F_{UI}(e.sid) \subseteq r.A \subseteq U$. □

C. System Invariants

I1 (Source Consistency). For any $s \in Reach(s_0)$ and any $e \in E(s)$, if Verify(e, R(s), T(s)) = accept, then there exists $r \in R(s)$ such that $e.ref = r.ref \wedge e.D_{sel} \subseteq r.A$.

I2 (Disclosure Minimality). For any $s \in Reach(s_0)$ and any $e \in E(s)$. $e.D_{sel} = F_{UI}(e.sid)$.

I3 (Session Locality). For any $s \in Reach(s_0)$, any $T \in T(s)$, and any $e \in E(s)$. $T.sid \neq e.sid \implies ValidToken(T, e) = 0$.

I4 (Channel Independence). For any $s \in Reach(s_0)$ and any $e \in E(s)$, the value of Verify(e, R(s), T(s)) depends only on (e, R(s), T(s), e.ch) and not on any event or state on the other channel.

D. Algebraic Properties of the Invariant System

Theorem 5.17 (Conjunctive Closure of Invariants): If predicates P_1 and P_2 are both invariants, then $P_1 \wedge P_2$ is also an invariant.

Proof. An invariant is defined as "holding for all $s \in Reach(s_0)$." If P_1 and P_2 hold for all reachable states, then $P_1 \wedge P_2$ also holds for all reachable states. □

Theorem 5.18 (Implicational Closure of Invariants): If P_1 is an invariant and $P_1 \implies P_2$, then P_2 is an invariant.

Proof. For any $s \in Reach(s_0)$, $P_1(s)$ holds; by the implication, $P_2(s)$ holds. □

Theorem 5.19 (I2 Implies the Inclusion Condition of I1): If for any $e \in E(s)$, $e.D_{sel} = F_{UI}(e.sid)$, and there exists $r \in R(s)$ such that $F_{UI}(e.sid) \subseteq r.A$, then I1 holds.

Proof. By Lemma 5.16, $F_{UI}(e.sid) \subseteq r.A$ and $e.D_{sel} = F_{UI}(e.sid)$, hence $e.D_{sel} \subseteq r.A$, i.e., condition (ii) of I1 holds. Condition (i) follows directly from the existence of r. □

Theorem 5.20 (Independence of I2 and I3): I2 and I3 are logically independent: there exist states satisfying I2 but not I3, and states satisfying I3 but not I2.

Proof. Construct examples. Satisfying I2 but not I3: the disclosure set of session sid_1 equals the interface output, but status token T_1 is incorrectly reused for verification of sid_2. Satisfying I3 but not I2: all tokens are correctly bound to sessions, but the disclosure set of some presentation event has fields additionally appended by the system. Hence the two are independent. □

E. Auxiliary Lemmas

Lemma 5.21 (Monotonicity of R): For any $s \to^* s'$, $R(s) \subseteq R(s')$.

Proof. By Definition 5.4, R grows only under issue(r) events and remains unchanged under show , token , and update events. Hence R is monotonically non-decreasing. □

Lemma 5.22 (Issuance Event Preserves Invariants): Let s satisfy I1 to I4, and $s' = \delta(s, issue(r))$. Then s' satisfies I1 to I4.

Proof. $R(s') = R(s) \cup \{r\}$, $K(s') \supseteq K(s)$, $T(s') = T(s)$, $E(s') = E(s)$. I1 is preserved by the monotonicity of R (Lemma 5.21); I2 and I3 are preserved because E and T are unchanged; I4 is preserved by the input structure of Verify. □

Lemma 5.23 (Presentation Event Preserves Invariants): Let s satisfy I1 to I4, $s' = \delta(s, show(e))$, and Verify(e, R(s), T(s)) = accept. Then s' satisfies I1 to I4.

Proof. R, K, T are unchanged, and E gains e. For the new event e, I1 holds by Verify conditions (i)(ii); I2 holds by Lemma 5.14; I3 holds by Lemma 5.15; I4 is preserved by the input structure of Verify. Existing events satisfy the invariants by the induction hypothesis. □

Lemma 5.24 (Token Event Preserves Invariants): Let s satisfy I1 to I4, and $s' = \delta(s, token(T'))$. Then s' satisfies I1 to I4.

Proof. R, K, E are unchanged, and T gains T'. I1, I2, and I4 obviously hold. I3 requires that T' satisfy session locality for any $e \in E(s')$, which follows directly from the signature covering sid and non_V at T' generation. □

Lemma 5.25 (Status Change Event Preserves Invariants): Let s satisfy I1 to I4, and $s' = \delta(s, update(ref, bit))$. Then s' satisfies I1 to I4.

Proof. R updates the status bit, A is unchanged, so all constraints involving A (condition (ii) of I1) are unchanged. I2, I3, and I4 are unchanged. □

F. Main Theorem

Theorem 5.26 (Reachability Invariant Theorem): For all $s \in Reach(s_0)$, s satisfies I1, I2, I3, and I4.

Proof. Induction on the length n of reachable paths. Base case n = 0: $s_0 = (\emptyset, \emptyset, \emptyset, \langle\rangle)$, $E(s_0) = \langle\rangle$, no events, so I1 to I4 hold vacuously. Inductive step: Assume s_n satisfies I1 to I4, and consider $s_{n+1} = \delta(s_n, ev)$. By Lemmas 5.22 to 5.25, all four event types preserve I1 to I4. By the induction principle, all reachable states satisfy I1 to I4. □

G. Corollaries and Extensions

Theorem 5.27 (Cross-Channel Disclosure Consistency Theorem): For any $s \in Reach(s_0)$ and the same session identifier sid, if s contains a remote-channel presentation event e_{remote} and a proximity-channel presentation event e_{prox} such that $e_{remote}.sid = e_{prox}.sid = sid$, then

$$e_{remote}.D_{sel} = e_{prox}.D_{sel} = F_{UI}(sid)$$

Proof. By I2, $e_{remote}.D_{sel} = F_{UI}(e_{remote}.sid) = F_{UI}(sid)$; $e_{prox}.D_{sel} = F_{UI}(e_{prox}.sid) = F_{UI}(sid)$. Hence the two are equal. □

Theorem 5.28 (Monotonic Preservation of Invariants over Event Histories): If $s \in Reach(s_0)$ satisfies I1 to I4, then for any event sequence σ, $s' = \delta^*(s, \sigma)$ satisfies I1 to I4.

Proof. By Lemmas 5.22 to 5.25, each event type preserves I1 to I4. Induction on the length of σ yields the result. □

Theorem 5.29 (Mutual Information Characterization of Unlinkability): Model the holder identity as a random variable H, the session identifier as a random variable SID, and the status token as a random variable TK. If TK's value is determined only by (ref, $status_{bit}$, non_V, sid, t), and H is conditionally independent of (ref, non_V, sid, t) given $status_{bit}$, then

$$I(H; \mathrm{TK}|\mathrm{SID}) = 0$$

where $I(\cdot;\cdot|\cdot)$ is conditional mutual information.

Proof. TK is a deterministic function of (ref, $\mathrm{status_{bit}}$, $\mathrm{non_V}$, sid, t). By the conditional independence assumption, H and TK are conditionally independent given SID. By the definition of conditional mutual information, I(H; TK | SID) = 0. □

Corollary 5.30 (Weak Form of Unlinkability): Under the conditions of Theorem 5.29, the status tokens ($\mathrm{TK_1}$, $\mathrm{TK_2}$) corresponding to any two different sessions carry no information about H:

$$I(H; \mathrm{TK_1}, \mathrm{TK_2}|\mathrm{SID_1} \neq \mathrm{SID_2}) = 0$$

Proof. By Theorem 5.29 and the chain rule for conditional mutual information. □

Theorem 5.31 (Discriminative Power of the Verification Predicate): For any e ∈ E(s), Verify(e, R(s), T(s)) = accept implies $\mathrm{e.D_{sel}} \subseteq$ r.A and T.sid = e.sid. That is, the verification predicate has complete discriminative power for the two types of anomalies: "disclosure overreach" and "token mismatch."

Proof. Verify condition (ii) directly checks $\mathrm{e.D_{sel}} \subseteq$ r.A; condition (iii) directly checks T.sid = e.sid. If either condition fails, Verify returns reject. Hence it has complete discriminative power for the two types of anomalies. □

VI. Formalization of Feasibility Boundaries

A. Feasibility Decision Function

Specification consistency does not imply deployment feasibility. This section formalizes feasibility as a conjunction of constraint predicates and proves structural properties of the feasible region.

Let the system implementation be sys and the target jurisdiction be J. Define the feasibility decision function

$$F(sys, J) = 1 \Longleftrightarrow \wedge_{i=1}^{6} \mathrm{C_i}(sys, J) = 1$$

where $\mathrm{C_i}$ is the i-th constraint predicate. External parameters are denoted θ(J) = (θ_1, θ_2, θ_3, ρ), and system internal parameters are denoted sys = (τ, log, δ).

B. Formalization of Constraint Predicates

$\mathrm{C_1}$ **(Credential Identity).** $\mathrm{C_1} = \theta_1$, where θ_1(J) ∈ {0,1} indicates whether J recognizes two credentials with the same ref as the same document.

$\mathrm{C_2}$ **(Format Compliance).** $\mathrm{C_2} = \neg\theta_2 \vee \delta$, where θ_2(J) ∈ {0,1} indicates whether J mandates 18013-5 mDL profile compliance, and δ ∈ {0,1} indicates whether J recognizes the relationship "proximity verification delegates to system-level mDL, and this system serves only as a data source."

$\mathrm{C_3}$ **(Status Management Equivalence).** $\mathrm{C_3} = \theta_3$, where θ_3(J) ∈ {0,1} indicates whether J recognizes that "presentation within the token validity period is equivalent to real-time lookup."

$\mathrm{C_4}$ **(Data Protection).** $\mathrm{C_4} = [\tau \leq \tau_{\max}(\mathrm{J})]$, where τ is the ref rotation period and $\tau_{\max}$(J) is the maximum reference lifetime allowed by J.

$\mathrm{C_5}$ **(Mutual Recognition Requirement).** $\mathrm{C_5} = \neg\rho$, where ρ(J) ∈ {0,1} indicates whether J requires cross- jurisdiction mutual recognition.

$\mathrm{C_6}$ **(Issuer Audit).** $\mathrm{C_6} = [\mathrm{audit_{ok}}(\log)]$, where $\mathrm{audit_{ok}}$ is the predicate that the log passes audit.

C. Feasible Region Characterization Theorem

Theorem 6.1 (Feasible Region Characterization Theorem): F(sys, J) = 1 if and only if

$$\begin{aligned} &\theta_1(J) = 1 \\ &\wedge\, \theta_3(J) = 1 \\ &\wedge\, \rho(J) = 0 \\ &\wedge\, (\theta_2(J) = 0 \vee \delta = 1) \\ &\wedge\, (\tau \leq \tau_{\max}(J)) \\ &\wedge\, \mathrm{audit_{ok}}(log) \end{aligned}$$

Proof. By the definitions in Section VI-B, $\mathrm{F = C_1 \wedge C_2 \wedge C_3 \wedge C_4 \wedge C_5 \wedge C_6}$. Expanding each constraint and rearranging using the commutativity of ∧ yields the required form. □

Corollary 6.2 (External Subset of the Feasible Region): Define

$$\begin{aligned} \mathrm{D_{ext}} = \{J : &\theta_1(J) = 1 \\ &\wedge\, \theta_3(J) = 1 \\ &\wedge\, \rho(J) = 0 \\ &\wedge\, (\theta_2(J) = 0 \vee \delta = 1)\} \end{aligned}$$

Then F(sys, J) = 1 implies $\mathrm{J} \in \mathrm{D_{ext}}$.

Proof. By Theorem 6.1, F = 1 requires $\theta_1 \wedge \theta_3 \wedge \rho = 0 \wedge (\neg\theta_2 \vee \delta)$, i.e., $\mathrm{J} \in \mathrm{D_{ext}}$. □

Corollary 6.3 (Internal Feasible Region): Define $\mathrm{D_{int}}$(J) = {(τ, log) : $\tau \leq \tau$_max(J) ∧ $\mathrm{audit_{ok}}$(log)}. Then for any $\mathrm{J} \in \mathrm{D_{ext}}$, there exists ($\tau$, log) ∈ $\mathrm{D_{int}}$(J) such that F(sys, J) = 1.

Proof. By Theorem 6.1, when $\mathrm{J} \in \mathrm{D_{ext}}$, F = 1 if and only if ($\tau$, log) ∈ $\mathrm{D_{int}}$(J). Hence any (τ, log) satisfying the conditions works. □

D. Structural Properties of the Feasible Region

Theorem 6.4 (Decomposability of the Feasible Region): The feasible region can be decomposed into the Cartesian product of the external feasible region and the internal feasible region:

$$\begin{aligned} (sys, J) : F(sys, J) = 1 = \{(sys, J) : J \in \mathrm{D_{ext}} \\ \wedge\, (\tau, log) \in \mathrm{D_{int}}(J)\} \end{aligned}$$

Proof. Direct combination of Theorem 6.1 with Corollaries 6.2 and 6.3. □

Theorem 6.5 (External Parameter Dominance): If there exist $\mathrm{J_1}$ and $\mathrm{J_2}$ such that $\theta(\mathrm{J_1}) = \theta(\mathrm{J_2})$ and $\rho(\mathrm{J_1}) = \rho(\mathrm{J_2})$, then for any sys, F(sys, $\mathrm{J_1}$) = F(sys, $\mathrm{J_2}$).

Proof. By Theorem 6.1, F's dependence on J is entirely through θ(J) and ρ(J); when these are equal, F takes the same value on $\mathrm{J_1}$ and $\mathrm{J_2}$. □

Theorem 6.6 (Monotonicity of F with Respect to τ): Fixing other parameters, if $\tau_1 \leq \tau_2$, then F(τ_1, log) ≥ F(τ_2, log).

Proof. $C_4 = [\tau \le \tau_{max}]$. When $\tau_1 \le \tau_2$, if $\tau_2 \le \tau_{max}$ then $\tau_1 \le \tau_{max}$, so $C_4(\tau_1) \ge C_4(\tau_2)$. Other constraints are independent of τ. Hence $F(\tau_1) \ge F(\tau_2)$. □

Theorem 6.7 (Monotonicity of F with Respect to log): If $\log_1 \subseteq \log_2$ (information inclusion), then $F(\tau, \log_1) \le F(\tau, \log_2)$.

Proof. $audit_{ok}$ is monotonic with respect to log information: if $\log_1$ passes audit, then $\log_2$ containing $\log_1$ should also pass audit. Hence $C_6(\log_1) \le C_6(\log_2)$. Other constraints are independent of log. □

Theorem 6.8 (Lipschitz Property of F with Respect to τ): If τ is treated as a continuous variable, F changes as a step function in τ, and its Lipschitz constant under discretization does not exceed 1.

Proof. C_4 is the indicator function $[\tau \le \tau_{max}]$, whose value jumps from 1 to 0 at $\tau = \tau_{max}$. Hence the difference of F with respect to τ is a step, and the Lipschitz constant under discretization does not exceed 1. □

Theorem 6.9 (Parameter Sensitivity of the Feasible Region): If θ_1 changes from 1 to 0, then for any sys, F changes from 1 to 0; if ρ changes from 0 to 1, then for any sys, F changes from 1 to 0. That is, θ_1 and ρ are necessary condition parameters of the feasible region.

Proof. By Theorem 6.1, $\theta_1 = 0$ directly makes $F = 0$; $\rho = 1$ directly makes $C_5 = 0$, hence $F = 0$. □

E. Optimization of Deployment Cost

Definition 6.10 (Deployment Cost Function): Let Cost: $D_{int} \to \mathbb{R}_\{\ge 0\}$ be a deployment cost function satisfying: (a) Cost is continuous on D_{int}; (b) as $(\tau, \log)$ approaches the boundary of D_{int}, $Cost(\tau, \log) \to +\infty$.

Theorem 6.11 (Existence of Optimal Deployment): If $D_{int}(J)$ is non-empty and compact, and Cost satisfies Definition 6.10, then there exists $(\tau, \log) \in D_{int}(J)$ such that $Cost(\tau, \log) = \inf$ Cost, and $F(sys^*, J) = 1$.

Proof. By the continuity and boundary divergence of Cost, Cost attains its infimum on the compact set $D_{int}(J)$ (Weierstrass extreme value theorem). By Corollary 6.3, any $(\tau, \log) \in D_{int}(J)$ makes $F = 1$. Theorem 6.12 (Convexity of Optimal Cost). If Cost is convex on D_{int} and D_{int} is convex, then the optimal solution set is convex. □

Proof. Standard result in convex optimization: the set of infimum points of a convex function on a convex set is convex. □

Theorem 6.12 (Fixed Point Existence): Define the deployment morphology operator Π: $D_{ext} \times D_{int} \to D_{ext} \times D_{int}$ as

$$\Pi(J, \tau, log) = (J', \tau', log')$$

where J' is the jurisdiction parameter set after the system feeds back to jurisdiction J under given $(\tau, \log)$ and re-determines feasibility, and τ', log' are the optimal system parameters determined under J' by Theorem 6.11. If Π is continuous on a compact convex set and maps the compact convex set into itself, then by Brouwer's fixed point theorem, Π has a fixed point $(J, \tau, \log^*)$.

Proof. Standard application of Brouwer's fixed point theorem. □

Corollary 6.13 (Meaning of the Fixed Point): If $(J, \tau, \log^*)$ is a fixed point of Π, then the deployment morphology is stable under the given jurisdiction rules: further optimization of system parameters does not change the feasibility decision, and further adjustment of jurisdiction parameters does not change the optimal system parameters.

F. Parameter Space Discussion

The values of external parameters θ_1, θ_3, and ρ are determined by the jurisdiction's regulatory environment and cannot be known at the design stage. By Theorem 6.4 (decomposability), a reasonable engineering practice is: before deployment, determine the values of $(\theta_1, \theta_3, \rho, \theta_2, \delta)$ for each target jurisdiction, decide $J \in D_{ext}$ accordingly, and then solve the optimal deployment cost problem of Theorem 6.11 on $D_{int}(J)$.

By Theorems 6.6 and 6.7, there is a trade-off in choosing system parameters τ and log: a shorter τ makes C_4 easier to satisfy but increases rotation burden; a more detailed log makes C_6 easier to satisfy but increases storage and audit costs. This trade-off does not change the feasible region D, only affects deployment cost on D, and belongs to engineering optimization, with the existence of an optimal solution guaranteed by the formalization of Theorem 6.11.

VII. Overview of Cross-Domain Applicability

The criteria in Section III-A define the applicability domain of the design. The digital driver's license is a typical instance satisfying P1 to P5, but not the only one. By credential attribute characteristics, this paper retains three scenarios with relatively high matching to the design: identity and qualification proof (digital identity cards, professional licenses), finance and credit (KYC credentials, income and tax certificates, insurance credentials), and assets and permissions (vehicle registration, access control permissions).

Appendix A provides item-by-item checks for the three scenarios, typical partitioning of disclosure sets, scenario differences in status timeliness, mappings from system invariants to scenarios, and several inapplicability counterexamples. The core conclusion is: the value of the design lies not in serving a particular credential type, but in providing a general method for maintaining source traceability, disclosure minimization, status timeliness, and channel independence in an ecosystem with fragmented formats; the digital driver's license is the first instance of this method, but not the only one.

VIII. Discussion

A. Boundaries of Formal Arguments

Section V argues that "under S1 to S4, the scheme satisfies the system invariants and their corollaries." The arguments do not cover three situations. First, scope exclusions: cross-format holder equivalence and trust anchor mapping are not modeled, and the model does not claim their validity through any theorem. Second, engineering risks: side channels, key management failures, implementation defects, etc., are outside the system specification model. Third, standards maturity risk: SD-JWT

VC is still a draft, and its final form may differ from current assumptions; the two premises "placeholder indistinguishability" and "disclosure set uniquely determined by the interface" need to be rechecked against the final standard.

B. Engineering Implementation Considerations

Co-generation requires the issuer to maintain two signature processes, JWS and COSE. In engineering, the same attribute source and the same ref computation logic can be reused, with the signature layers implemented separately. For key management, the two sets of keys should be independently generated and independently rotated to avoid single-point failure spreading to both formats. The transaction log should record the ref and timestamp of each co-generation as evidence of satisfying C_6.

The proximity channel reuses OpenID4VP proximity; its engineering advantage is reusing the session management and verification code of the remote channel, reducing implementation surface; its limitation is that verification devices need to support this mode, and existing 18013-5 terminals do not necessarily support it. This limitation has been stated in Section III-D and is a capability boundary rather than an implementation defect.

C. Rationality of the Design Choice

This paper positions the "format coexistence" problem at the issuance stage rather than the presentation stage. The rationality of this choice is directly supported by the theorems in Section V: solving it at the issuance stage makes source consistency a monotonic invariant of reachable states (Theorem 5.26), makes cross-channel disclosure consistency degenerate into an identity (Theorem 5.27), and makes session locality directly excludable by the conjunctive conditions of the verification predicate (Lemma 5.15). None of the three properties relies on any intermediate component's self-declaration, and thus they can be directly checked locally by the verifier. This is the direct benefit of the design choice, not a comparative advantage over some alternative.

D. Limitations

This paper has four limitations. First, the formal model does not cover cross-format holder equivalence; if a business scenario requires this property, a binding mechanism must be designed outside the scheme, which this paper does not provide. Second, the feasibility analysis treats jurisdiction parameters as binary variables; actual regulatory environments may be more complex, and the model is a simplification of reality. Third, this paper does not quantitatively evaluate performance and availability; the impact of co-generation on issuer throughput needs to be measured in specific deployments. Fourth, the scenario analysis in Appendix A is based on qualitative matching of criteria and does not quantitatively rank the matching degree of various scenarios; determining scenario priorities requires market-specific research.

IX. Conclusion

This paper proposes and formally evaluates a co-generation and channel orthogonalization design that natively generates at the issuance stage and introduces no format intermediary at the presentation stage. Formal modeling of system specifications shows that, under the three system assumptions of issuance record integrity, reference function determinism, and device key exclusivity, the design satisfies system invariants including source consistency, disclosure minimality, session locality, and channel independence, with cross-channel disclosure consistency degenerating into a direct corollary of disclosure minimality (Theorem 5.27). The main theorem 5.26, by induction over the reachable state set, provides a rigorous argument that the invariants hold for all reachable states; Theorems 5.29 and Corollary 5.30 provide a mutual information characterization of unlinkability; Theorem 5.31 provides complete discriminative power of the verification predicate for two types of anomalies.

The feasibility analysis formalizes the deployment decision as a conjunction of constraint predicates. Theorem 6.1 provides an exact characterization of the feasible region; Theorem 6.4 proves that the feasible region decomposes into the Cartesian product of external and internal feasible regions; Theorems 6.6, 6.7, and 6.8 provide monotonicity and Lipschitz properties of F with respect to system parameters; Theorem 6.9 provides parameter sensitivity; Theorems 6.11, 6.12, and 6.13 provide existence, convexity, and fixed point existence for optimal deployment cost solutions. This series of theorems transforms "whether deployment is possible" from a design discussion into parameter determination and an optimization problem, providing an auditable basis for engineering decisions.

The applicability of the design is defined by a set of applicability criteria. The digital driver's license is the first complete instance, and Appendix A shows that identity and qualification proof, finance and credit, and assets and permissions scenarios apply to varying degrees, while also providing several inapplicability counterexamples. In summary, the design holds at the specification consistency level and is conditionally feasible at the deployment level; its boundaries can be precisely modeled but cannot be eliminated by technical solutions. The value of a technical solution lies not in eliminating boundaries, but in making boundaries computable, so that subsequent decisions are based on an accurate understanding of the facts.

Appendix A
Cross-Domain Applicability Analysis

A. From Driver's Licenses to General Credentials: Abstraction

The main text develops the digital driver's license as an instance, but the model in Section III and the invariants in Section V do not depend on the specific attributes of a driver's license. Replacing the driver's license with any credential type, as long as the credential simultaneously satisfies the applicability criteria in Section III-A, the model and invariants hold as-is. The digital driver's license satisfies P1 to P5 simultaneously and is therefore a typical instance. But the above criteria are not unique to driver's licenses–any credential that is "authoritatively issued, status-changing, requires minimal disclosure, dual-channel verification, and coexisting formats"

TABLE I
QUALITATIVE MATCHING OF APPLICATION SCENARIOS TO APPLICABILITY CRITERIA

Scenario Category	P1: Authoritative Source	P2: Status Change	P3: Minimal Disclosure	P4: Dual Channel	P5: Format Coexistence
Identity and Qualification Proof	Strong	Strong	Strong	Strong	Strong
Finance and Credit	Strong	Medium–Strong	Strong	Medium	Strong
Assets and Permissions	Medium	Medium	Medium	Strong	Medium

TABLE II
MAPPING OF SYSTEM INVARIANTS TO APPLICATION SCENARIOS

Invariant	Meaning in Driver's License Scenario	Typical Mapping in Other Scenarios
I1: Source Consistency	Every field seen by the verifier comes from some issuance.	Finance: KYC fields must come from a due-diligence issuance. Professional licenses: practice qualification must come from a license issuance.
I2: Disclosure Minimality	The user-confirmed field set is identical to the actual disclosure.	Finance: loan approval releases only an income range. Professional licenses: only the practice scope is released.
I3: Session Locality	The status token is limited to a single presentation session.	Professional-license inspection: each on-site inspection uses an independent session, and the last passed token cannot be reused. KYC: verification logs are retained locally, while the token itself is not persistent.
I4: Channel Independence	Remote and proximity verification are mutually independent.	Insurance: remote annual-inspection verification and on-site accident verification do not affect each other. Access control: remote visitor appointments and on-site card reading are mutually independent.

can apply the same method. The following retains three scenarios with relatively high matching to the design and analyzes them one by one.

B. Application Scenario Classification

1) Identity and Qualification Proof: **Digital Identity Card (eID).** The attribute source is the household registration authority, and the status is validity period and lost status. Remote verifiers are bank account opening, telecommunications real-name registration, and government online services; proximity verifiers are hotel check-in, airport security auxiliary verification, and government service windows. Typical partitioning of disclosure sets: hotels only need name and photo; banks need name, document number, and validity period. This scenario is isomorphic to driver's licenses and is the most direct extension.
Professional Licenses. Medical practitioner licenses, lawyer licenses, electrician certificates, welder certificates, special equipment operator certificates. The attribute source is the industry regulatory authority, and the status is valid, suspended practice, revoked, and expired. Remote verifiers are employer HR systems and industry association query platforms; proximity verifiers are regulatory inspectors scanning at practice sites. This scenario has higher status timeliness requirements than driver's licenses– presentation during suspension must be immediately identified. I3 (session locality) directly corresponds in this scenario to "each on-site inspection by an inspector uses an independent session, and tokens cannot be reused."

2) Finance and Credit: **KYC Credentials.** The attribute source is a bank or licensed KYC institution, and the status is valid, expired, and risk level changed. Remote verifiers are securities account opening, payment institution registration, and cross-border remittance; proximity verifiers are counter large-value transactions. The special feature of this scenario is that verifiers have retention obligations–banks must retain verification records for regulatory inspection. This creates tension with I3 (session locality): session locality requires tokens not to be reused across sessions, but retention obligations require verifiers to trace afterward. The design's approach is: the token itself is not persistent, but the verifier locally retains a log of "a certain session verification passed"; the retention period and format of the log are determined by regulatory rules and do not conflict with token unlinkability.
Income and Tax Certificates. The attribute source is the tax authority or employer, and the status is year and number of times issued. Remote verifiers are loan approval, visa applications, and rental review; proximity verifiers are government service windows. Disclosure minimality requirements: loan approval only needs an annual income range, not monthly details; rental review only needs whether income reaches a rent multiple, not the complete tax return.
Insurance Credentials. Auto insurance, travel insurance, property insurance. The attribute source is the insurance company, and the status is policy validity period, claim status, and surrender. Remote verifiers are annual inspection systems and claim verification systems; proximity verifiers are traffic accident on-site verification and claim on-site surveys. The auto insurance scenario is highly coupled with driver's licenses– traffic police may verify both the driver's license and compulsory auto insurance during roadside checks. The two credentials can be presented separately or combined in the same session, but the design does not require them to be bound.

3) Assets and Permissions: **Vehicle Registration Certificates.** The attribute source is the vehicle management office, and the status is valid, mortgaged, seized, and scrapped. Remote verifiers are used-car trading platforms and insurance underwriting systems; proximity verifiers are roadside enforcement and parking lot management. This scenario belongs to the same traffic management domain as driver's licenses and can share the same issuer infrastructure, but the credential type is independent. Access Control and Campus Permissions. The attribute source is the enterprise or campus management party, and the status is validity period, permission level, and lost. Remote verifiers are visitor appointment systems; proximity verifiers are access card readers. The characteristic of this scenario is that the issuer is not a government agency; P1's authority is provided by the enterprise's own trust declaration, and the value of C_1 (credential identity) in the feasibility

decision is determined by inter-enterprise agreements rather than jurisdiction rules.

C. Mapping from Scenarios to the Model

1) Typical Verifiers for the Two Channels: Typical verifiers for the remote channel (OpenID4VP + SD-JWT VC) are organizations with system integration capabilities: bank account opening systems, government online services, and insurance underwriting systems. Their common features are: they have backend services, can initiate presentation requests, can verify JWS, and can query status.

Typical verifiers for the proximity channel (OpenID4VP proximity + mDoc) are terminals in face-to-face scenarios: traffic police handheld terminals, government service windows, regulatory inspection terminals, scenic area turnstiles, and access card readers. Their common features are: verification occurs at physical sites, may lack stable networks, and requires rapid results.

2) Scenario Differences in Status Management: Different scenarios have significantly different requirements for status timeliness, which directly affects the token validity window design:

High-timeliness scenarios (professional license suspension, driver's license suspension, insurance surrender): the token validity window should be set to minutes, and offline acceptance rules should be strictly limited. In such scenarios, I3 (session locality) is most valuable–each on-site inspection by an inspector uses an independent session, avoiding reuse of "last inspection passed."

Medium-timeliness scenarios (KYC risk level, insurance policy status): the token validity window can be set to hours, and offline acceptance rules can be moderately relaxed.

Low-timeliness scenarios (access permissions, long-term qualifications): status changes are infrequent, the token validity window can be set to days, and even long-term assertions of "status unchanged" can be accepted.

3) Typical Partitioning of Disclosure Sets: Different scenarios have different logics for partitioning disclosure sets. The design does not prescribe partitioning rules but provides partitioning capability:

Partitioning by verifier type (driver's licenses, identity cards): traffic police see all; rental car companies see permitted vehicle categories and status; hotels see name and photo.

Partitioning by attribute hierarchy (professional licenses): basic qualification and additional endorsements are layered; verifiers request the corresponding hierarchy by purpose.

Partitioning by sensitivity (finance and credit, assets and permissions): highly sensitive fields such as income details and permission levels do not enter any verifier's request scope by default unless there is a clear legal basis.

D. Item-by-Item Checks of Applicability Criteria and Counterexamples

1) Criteria Checks: The matching of the above scenarios to P1 to P5 can be summarized as:

The matching of P4 and P5 determines whether the design is "necessary" rather than "usable." If a scenario has only remote verification and no proximity verification (e.g., pure online services), then P4 does not hold, and the design degenerates to a single-channel SD-JWT VC scheme, making dual issuance unnecessary cost. If a scenario has only proximity verification and no remote verification (e.g., pure offline access control), then P4 also does not hold, and the design degenerates to a single-channel mDoc scheme.

2) Inapplicability Counterexamples: The following scenarios do not satisfy the criteria, and the design should not be applied:

Real-Time Location and Dynamic Balances. Attributes require continuous updates (e.g., ride-hailing vehicle location, account balance). The credential model assumes "attributes fixed at issuance and status changes after issuance," which is incompatible with continuously updating attributes. Such scenarios should use real-time API queries rather than credential presentation.

Strong Real-Time Payment Authorization. Payment authorization requires millisecond-level response and online status checks. The "short-lived + offline acceptance" design of status tokens is neither necessary nor fast enough in this scenario. Such scenarios should use online authorization protocols rather than credential presentation.

Single-Verifier, Single-Channel Internal Systems. Enterprise internal permission systems and single- merchant internal systems do not have the P4 and P5 constraints; dual issuance and channel orthogonalization are over-engineering.

Scenarios Requiring Cross-Jurisdiction Mutual Recognition with No Negotiation. The scope exclusions in Section IV-C directly cause C_5 (mutual recognition requirement) to take value 0, making the design infeasible in such scenarios. If the business must have cross-border mutual recognition, it should use a native mDL scheme compliant with the IACA trust chain rather than the design in this paper.

Scenarios with Unclear Attribute Sources or Multi-Source Conflicts. If the same attribute has multiple mutually contradictory authoritative sources (e.g., different departments have inconsistent records of the same enterprise qualification), P1 does not hold. Although I1 (source consistency) of the design still holds (each credential can be traced to its issuance record), the verifier cannot determine which credential to trust. Such scenarios require additional source priority rules beyond the scope of this paper.

E. Mapping from Invariants to Scenarios

Key considerations for each invariant in different scenarios are as follows:

F. Summary

The applicability of the design in this paper is not limited to digital driver's licenses. Any verifiable credential satisfying P1 to P5–identity proof, professional licenses, financial KYC, asset registration, access permissions–can apply the same method, and the system invariants in Section V and the feasible region determination in Section VI hold as-is. The value of the design lies not in serving a particular credential type, but in providing a general method for maintaining source traceability, disclosure minimization, status timeliness, and channel independence in an ecosystem with fragmented formats. The digital driver's license is the first instance of this method, but not the only one. The matching of P4 and P5 determines whether the design is necessary; scenarios not satisfying the criteria should use simpler single-channel schemes rather than treating the design in this paper as the default choice.